RANJEET ANAND KUMBHAR

kumbhar.ranjeet01@gmail.com | (+91) 7350317656 | GitHub/Ranjeetkumbhar01 | LinkedIn/ranjeet-kumbhar | Pune, IN

Professional Summary

Software Engineering graduate student with strong foundations in Data Structures, Algorithms, and Machine Learning, with a keen research interest in applying advanced AI/ML techniques to real-world engineering problems. Qualified GATE 2024 (Data Science & AI) with strong problem-solving skills. Effective communicator and team player with a passion for technological innovation.

Education

Thapar Institute of Engineering & Technology, Patiala — *2024 – 2026*
Master of Engineering in Computer Science and Engineering — **CGPA: 9.0/10**

Savitribai Phule Pune University, Pune — *2020 - 2024*
Bachelor of Engineering in Information Technology (Minor: AI/ML) — **CGPA: 9.36/10**

Research & Publications

- Sensitivity-Driven Deep Learning Model for Tribological Prediction in Al7075/B_4C Nanocomposites, Journal submission under review, 2026. (Paper Link)
- Tribo-Informatics Method for Predictive Modeling of Friction and Wear Mechanisms in Al7075/B_4C Metal Matrix Composites, Journal submission under review, 2026. (Paper Link)
- A Novel Hybrid Controller for Sit-to-Stand Assistance using a Simscape Model based Lower-Limb Exoskeleton, ICMIC 2026 (Paper Link)

Work Experience

Thapar Institute of Engineering & Technology — Patiala, IN
Research Intern — ***Aug 2025 – May 2026***

- Spearheaded development of an AI-based system for Lower-Limb Exoskeletons to achieve stable gait patterns
- Authored multiple research papers on AI-driven frameworks and exoskeleton-based rehabilitation
- Built ML/DL based framework for tribological analysis using experimental datasets

College of Military Engineering (Indian Army) — Pune, IN
Research Internship — ***Dec 2022 – Mar 2023***

- Designed and developed a blockchain-based secure file sharing system in a private distributed LAN environment
- Implemented robust security measures to ensure secure file sharing
- Secured file sharing platform within a LAN environment

Projects

A Deep Learning–Based Controller for Lower-Limb Exoskeleton Assistance (M.E. Thesis - Ongoing)

- Designing an intelligent control framework for lower-limb exoskeletons to assist patients in achieving a stable gait pattern
- Developing deep learning–based predictive controllers for joint trajectory estimation
- Tech Stack: MATLAB/Simulink, Control Systems, Robotics, Artificial Intelligence

CataractAI: Deep-Learning Based Cataract Prediction System (GitHub)

- Developed a medical AI system leveraging computer vision for fundus image analysis, achieving over 90% accuracy
- Implemented EfficientNetB0 architecture with transfer learning from ImageNet
- Tech Stack: Flask, MySQL, TensorFlow, Deep Learning, Computer Vision

CarPoint: Car Model Recommendation System (GitHub)

- Developed a personalized recommendation engine for automobiles using Random Forest & ANN
- Enabled users to find their ideal car model based on their preferences and needs
- Tech Stack: Random Forest, Artificial Neural Network, TensorFlow, Django, Pandas, NumPy, Matplotlib

Skills & Interests

Technical/Tools: AWS, C, C++, Python, SQL, HTML/CSS, Bootstrap, Django, Git
Machine Learning: Deep Learning, NLP, Computer Vision, Statistical Analysis
Libraries: TensorFlow, Hugging Face, OpenCV, Matplotlib, Seaborn, Pandas, Scikit-learn, NumPy
Interests: Riding Bicycle, Exploring Financial Markets, Playing Games, Practicing Meditation